

House Price Prediction

Using Machine Learning

Presented by: **B Mohammed Afraz Hussain Basha**

Data Science | **Summer Internship Mini Project**

Problem Statement

Challenge

House prices vary significantly based on a large number of factors location, quality, size, amenities, and market trends making manual valuation inaccurate and time-consuming.

Goal

Build a machine learning model that can accurately predict house sale prices using historical data and key property features.

Impact

Provide buyers, sellers, and real estate agents with data-driven pricing estimates to support better decision-making.

Approach

Exploratory data analysis → data cleaning → feature selection → Linear Regression modeling → evaluation using MAE, RMSE, and R^2 Score.

Dataset Overview

Source

Kaggle

House Prices Dataset

Records

1,460

Training rows

Features

81

Columns

Target

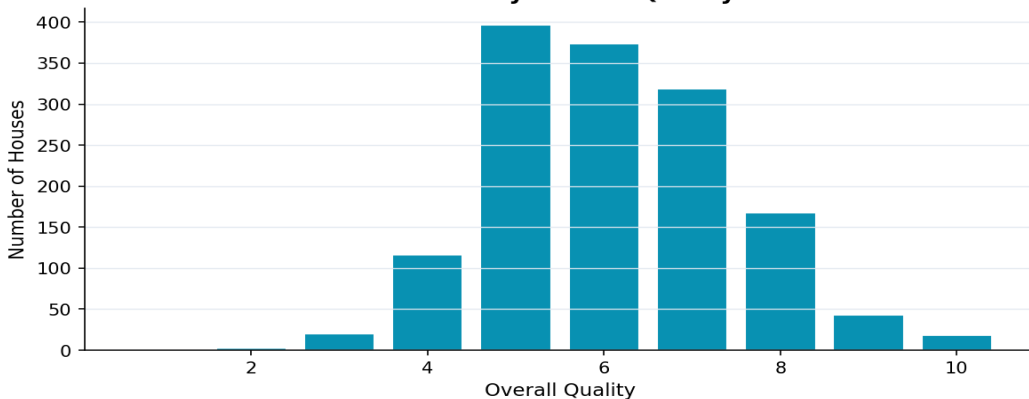
SalePrice

Regression output

Key Features Used in Modeling

- OverallQual
- GrLivArea
- GarageCars
- GarageArea
- TotalBsmtSF

Houses by Overall Quality



Data Cleaning

01

Missing Value Detection

Identified columns with missing values. Columns with >40% missing data (e.g., PoolQC, Alley, Fence) were flagged for removal.

02

Column Removal

Dropped columns with excessive missing values to avoid noise and overfitting in the model.

03

Numerical Imputation

Missing numerical values filled with the column median — robust to outliers unlike mean imputation.

04

Categorical Imputation

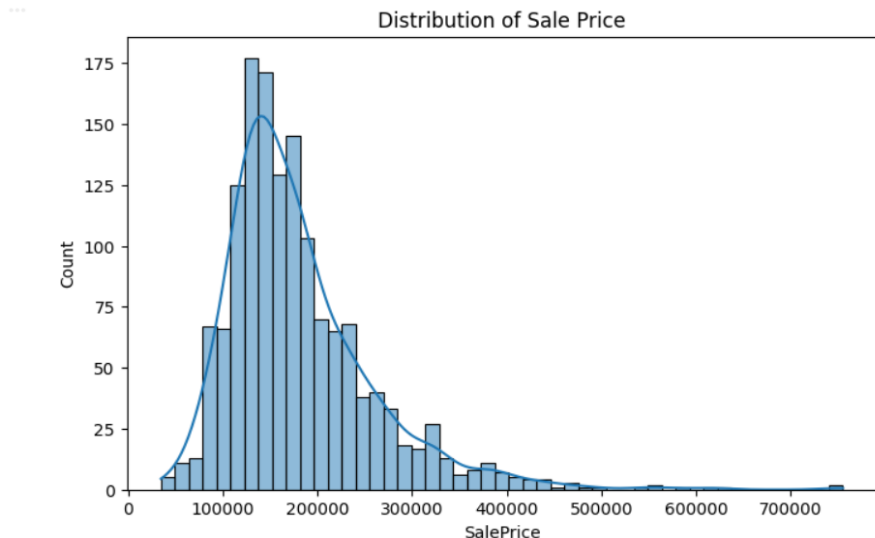
Missing categorical values filled with the column mode (most frequent value) to maintain feature distribution.

Missing Value Summary

Column	Missing %	Action
PoolQC	99.5%	Dropped
MiscFeature	96.3%	Dropped
Alley	93.8%	Dropped
Fence	80.8%	Dropped
LotFrontage	17.7%	Median
GarageType	5.5%	Mode
BsmtQual	2.5%	Mode

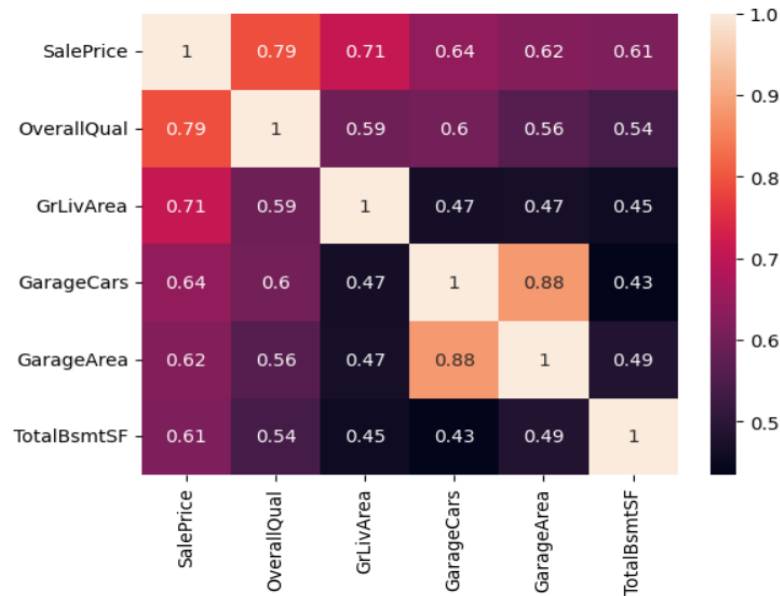
Exploratory Data Analysis

Sale Price Distribution

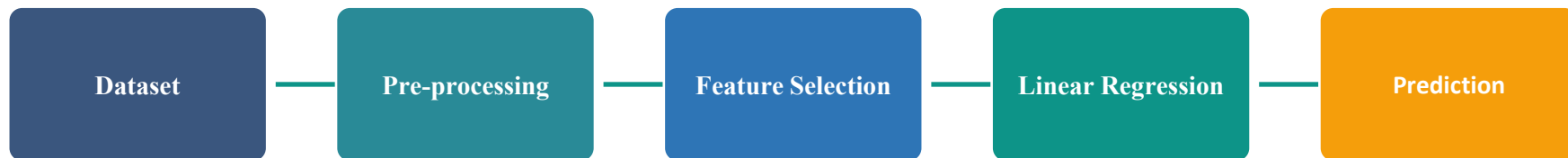


SalePrice is right-skewed. OverallQual (0.79), GrLivArea (0.71), and GarageArea (0.62) show the strongest correlations.

Correlation Heatmap



Model Building



Pre-processing

- Label encoding for categorical variables
- Median imputation for numerical nulls
- Mode imputation for categorical nulls

Feature Selection

- Correlation analysis with SalePrice
- Selected Top 5 Features
 - OverallQual
 - GrLivArea
 - GarageCars
 - GarageArea
 - TotalBsmtSF
- Train/Test split: 80% / 20%

Linear Regression

- Ordinary Least Squares (OLS) method
- Finds best-fit hyperplane in n-dimensions
- Minimizes sum of squared residuals

Model Results

MAE

25,284

Mean Absolute Error

RMSE

39,979

Root Mean Squared Error

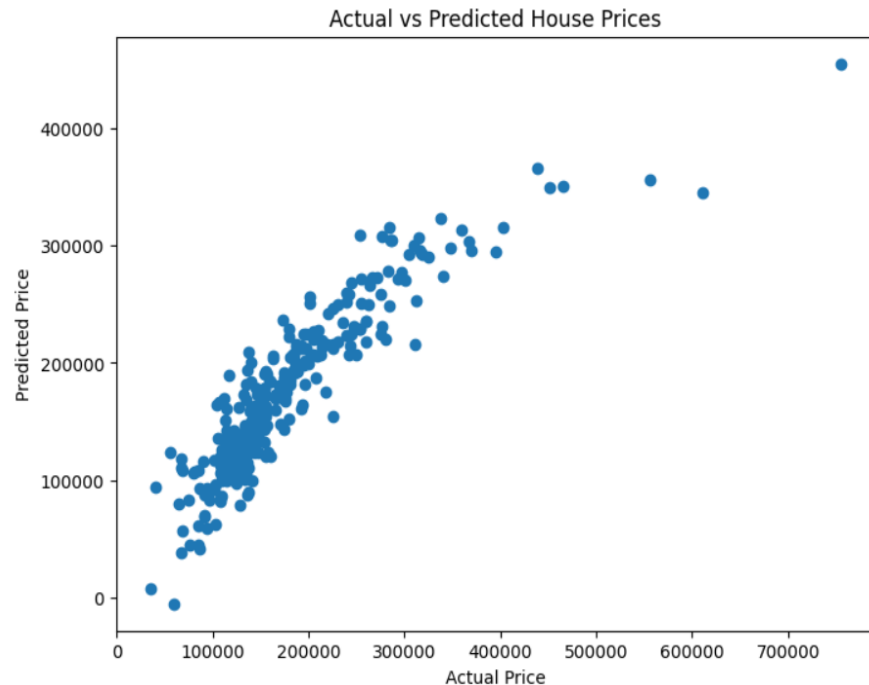
 R^2 Score**79.16%**

Model Accuracy

Performance Metrics

Metric	Value	Interpretation
MAE	\$25,284	Average prediction error
RMSE	\$39,979	Penalizes large errors
R^2 Score	79.16%	Variance explained

Actual vs Predicted Sale Price



Conclusion

- Model achieves 79.16% accuracy (R^2) — a strong baseline for house price prediction.
- Overall Quality (OverallQual) is the single strongest predictor of sale price.
- Living Area (GrLivArea) and Garage Area (GarageArea) significantly influence prices.
- Future scope: Random Forest and XGBoost expected to push accuracy

Thank You